LAB 3 | Hospital Appointment and Patient Triage

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Course: WORKFLOW MANAGEMENT

Camunda 8.8 | Java 17 | Spring Boot 3.5.6 | Synthetic data

Aim

Register a synthetic patient request, apply educational urgency flags, involve staff where necessary, and retrieve/book a mock appointment while supporting incident recovery.

Architecture and execution

1. The starter accepts patientRef, appointmentPreference, symptomsSummary, urgentIndicator, priorityIndicator and slotScenario. All records are fictional.

2. The hospital-register worker validates required fields, boolean indicators, supported preference and scenario. Invalid registration ends before triage.

3. HospitalTriage uses FIRST hit policy: urgentIndicator wins even when both flags are true; otherwise priorityIndicator selects Priority Review; both false select Routine.

4. Urgent cases go immediately to the human-assessment Tasklist task, without waiting for availability or a booking API. Completion records the human assessment outcome.

5. Priority requests go to staff review. Approve continues scheduling; Reject ends as Referred to staff. Routine requests go directly to availability.

6. The hospital-availability Java worker requests preliminary slots. The REST connector refreshes the slot list, and the availability gateway uses the fresh result.

7. With slots, the booking worker posts the first slot to /mock/bookings and records Booked. With no slots, a staff task records a follow-up action and ends as Staff follow-up required.

Rules, samples and assumptions

Hit policy: FIRST. The first matching rule returns one output.

Order	Condition	Output
1	urgentIndicator = true; priorityIndicator = any	Urgent Assessment
2	urgentIndicator = false; priorityIndicator = true	Priority Review
3	Both flags false	Routine

Urgent flag; priority flag	Expected result
false; false	Routine
false; true	Priority Review
true; false	Urgent Assessment
true; true	Urgent Assessment

Validation

The existing local suite passed 39 JUnit tests, including 19 standalone DMN evaluations, plus 20 mock HTTP checks. Standalone rule tests used Camunda 7.24; live workflow screenshots separately demonstrate Camunda 8 process execution. Logs are retained under evidence/.

Assumptions

This is not a clinical triage or emergency-care application. The symptoms text is stored for review, not interpreted by a medical model. Slot identifiers are synthetic sessions, not real appointments. Booking is an in-memory idempotent mock without cross-patient capacity enforcement. The demo account represents staff for the exercise; production access controls are outside scope.

Configuration and variables

README.md supplies startup/deployment commands. API_EXAMPLES.md documents REST mappings and request/response bodies. VARIABLE_DICTIONARY.md defines variable names, types and purposes. Editable models and all source code are included.

Observed runtime results

Live execution was performed on a Windows/Docker installation on 30-31 October 2026. Results below were transcribed from Operate/Tasklist screenshots and copied process variables. Fifteen required scenarios have observed terminal outcomes. Seven additional example scenarios are not claimed as live-tested. Screenshot times are shown as displayed on the test PC.

Case / input JSON	Expected	Observed / instance	Status
H1-routine.json	Booked	Booked 2251799813697163	Passed
H2-priority.json	Priority staff review; Booked after continuation	Priority staff review -> continuation -> Booked 2251799813716434	Passed
H3-urgent.json	Immediate human assessment; Human assessment recorded after completion	Human assessment recorded; staffActionRecorded=true 2251799813717405	Passed
H4-no-slots.json	No-slot staff task; Staff follow-up required after completion	No slots; staff review -> Staff follow-up required 2251799813717956	Passed
H5-scheduling-error.json	Availability worker incident; Booked after fault cleared and retry	Lab 3 HTTP 503 dashboard observed; user reported correction/retry; H5 completed Booked 2251799813723383	Passed with evidence limitation

Additional examples

H6-priority-refer, H7-invalid: supplied but not live-tested.

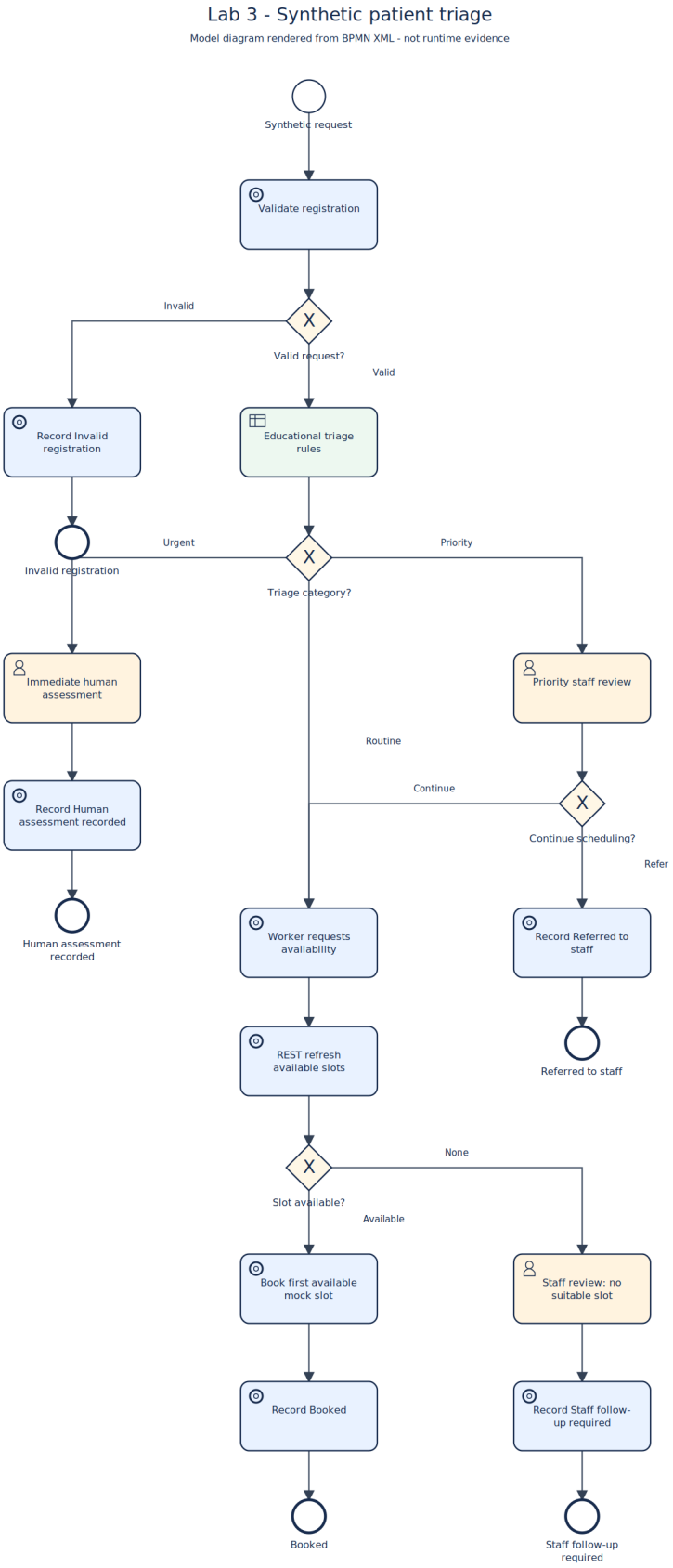
Evidence boundaries

The H5 incident capture shows a Lab 3 HTTP 503 error on the dashboard, without an instance key. Recovery is shown for H5 instance 2251799813723383. The supplied test sequence links the events, but the images do not independently establish a matching before/after key. Human task captures show forms or task lists; final outcomes are shown in Operate.

Recovery procedure

Enable the case-specific mock fault, start the example and inspect the incident in Operate. Clear that fault and retry the failed job in the same instance. The failed external service returns HTTP 503; worker/connector failure exhausts retries and creates an incident. Loan worker faults are injected in validation. H5 reaches the scheduling API first through the availability worker. No real services or payments are involved.

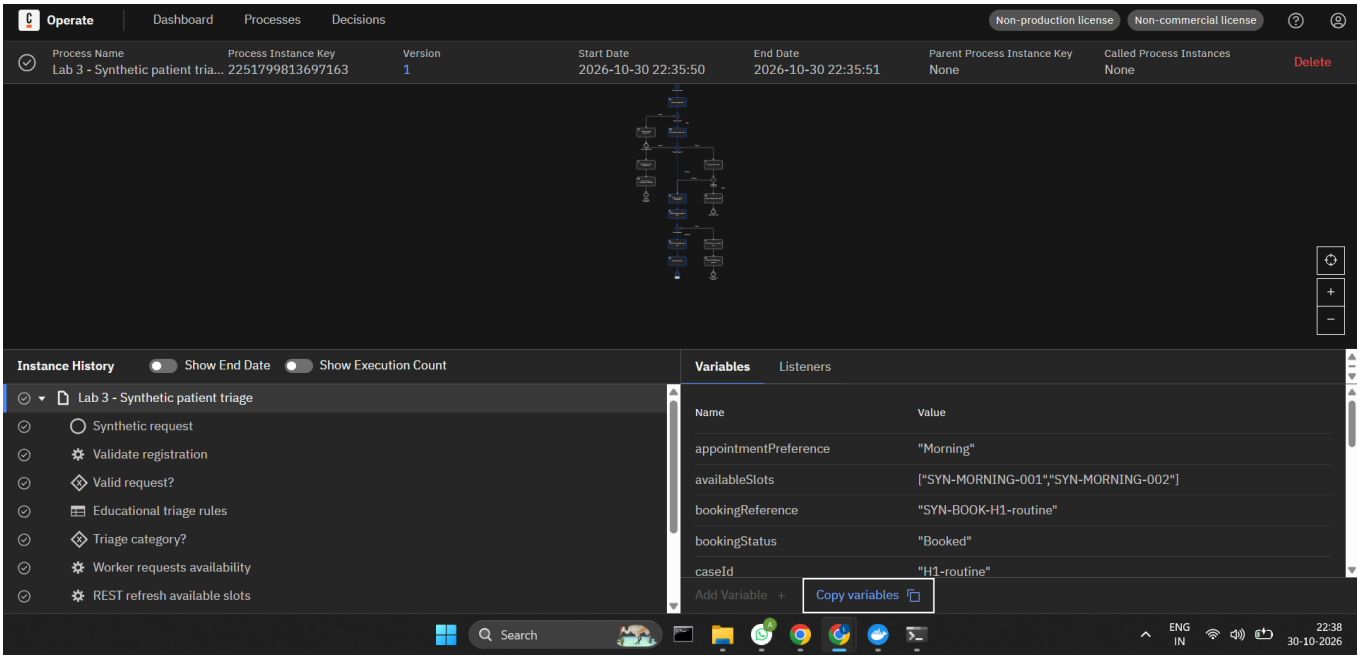
BPMN process design



Rendered from the supplied BPMN XML. Runtime captures follow; full-resolution original screenshots are in evidence/runtime/.

H1-booked

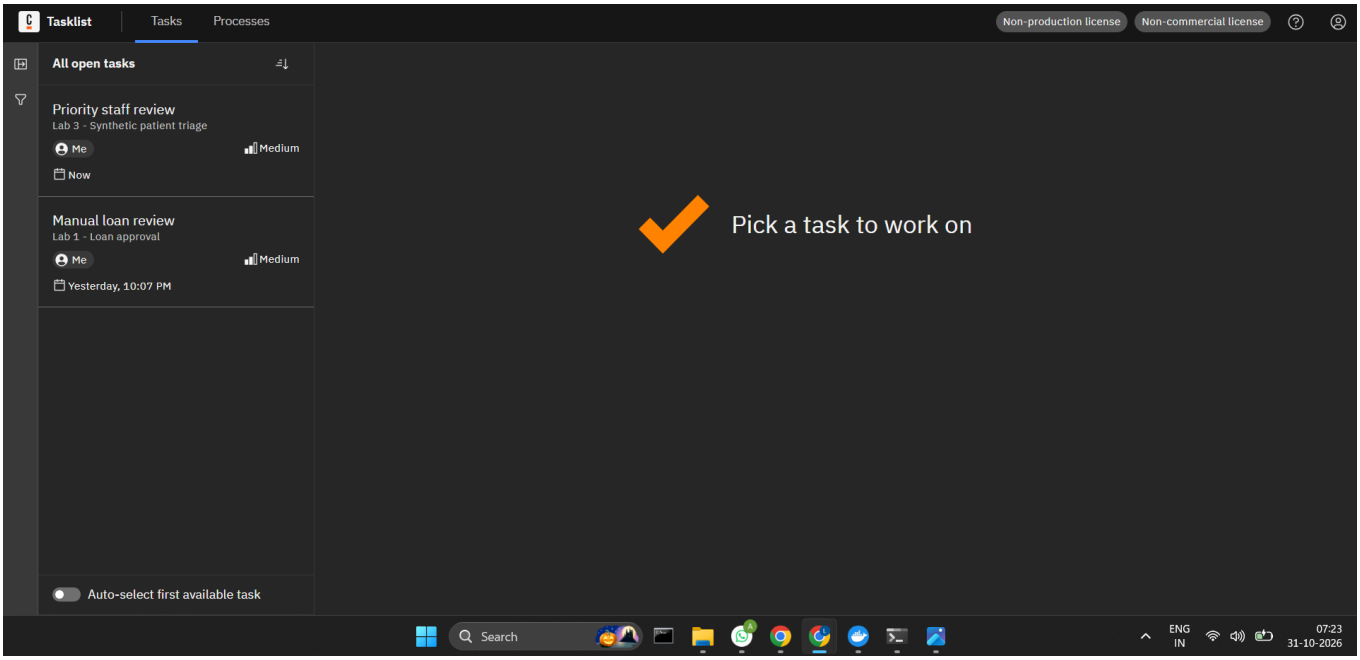
Routine case booked.



Full-resolution evidence: evidence/runtime/H1-booked.png

H2-task-list

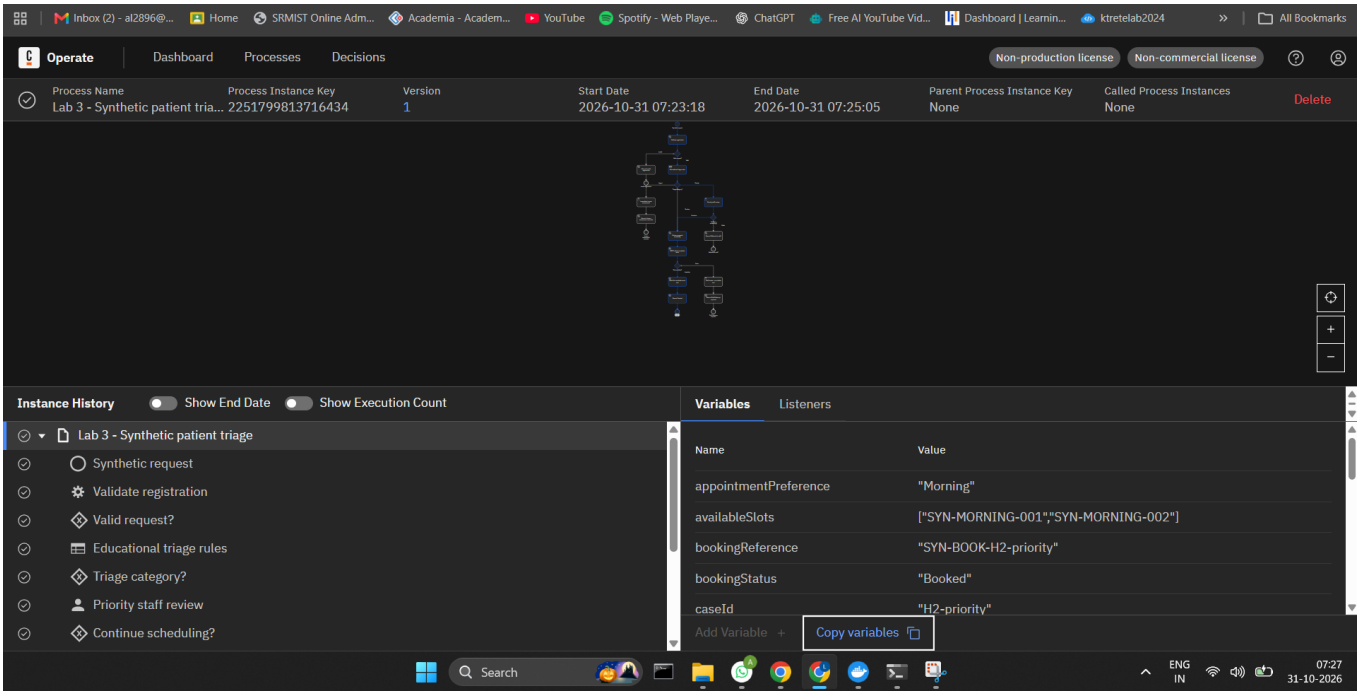
Priority review task listed in Tasklist.



Full-resolution evidence: evidence/runtime/H2-task-list.png

H2-booked

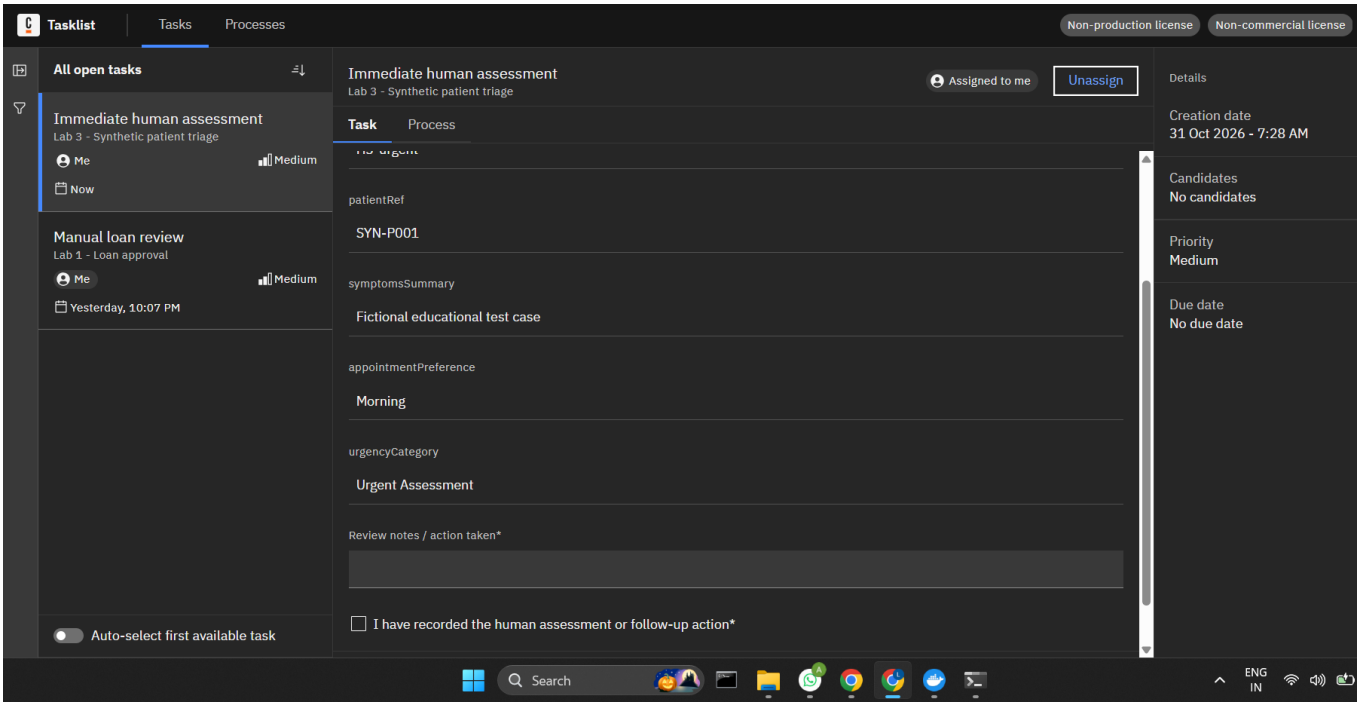
Priority staff review executed; Booked result.



Full-resolution evidence: evidence/runtime/H2-booked.png

H3-task-form

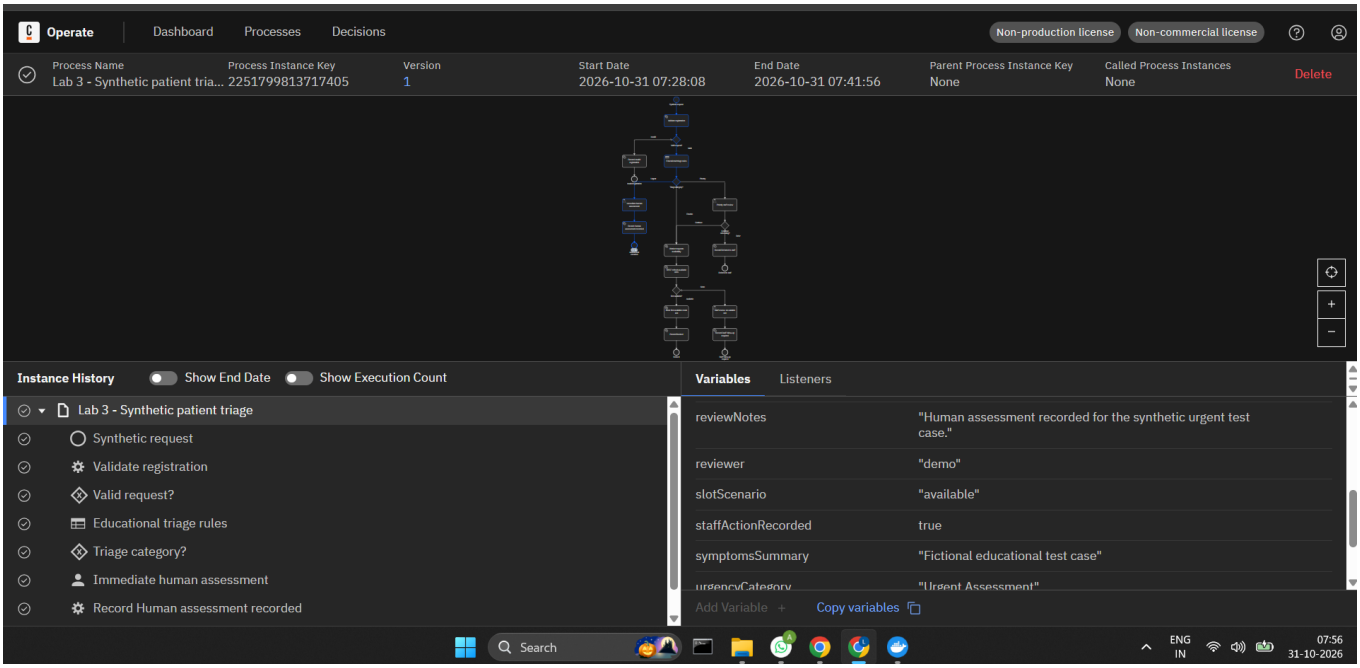
Immediate human assessment form before completion; not a submitted form.



Full-resolution evidence: evidence/runtime/H3-task-form.png

H3-completed

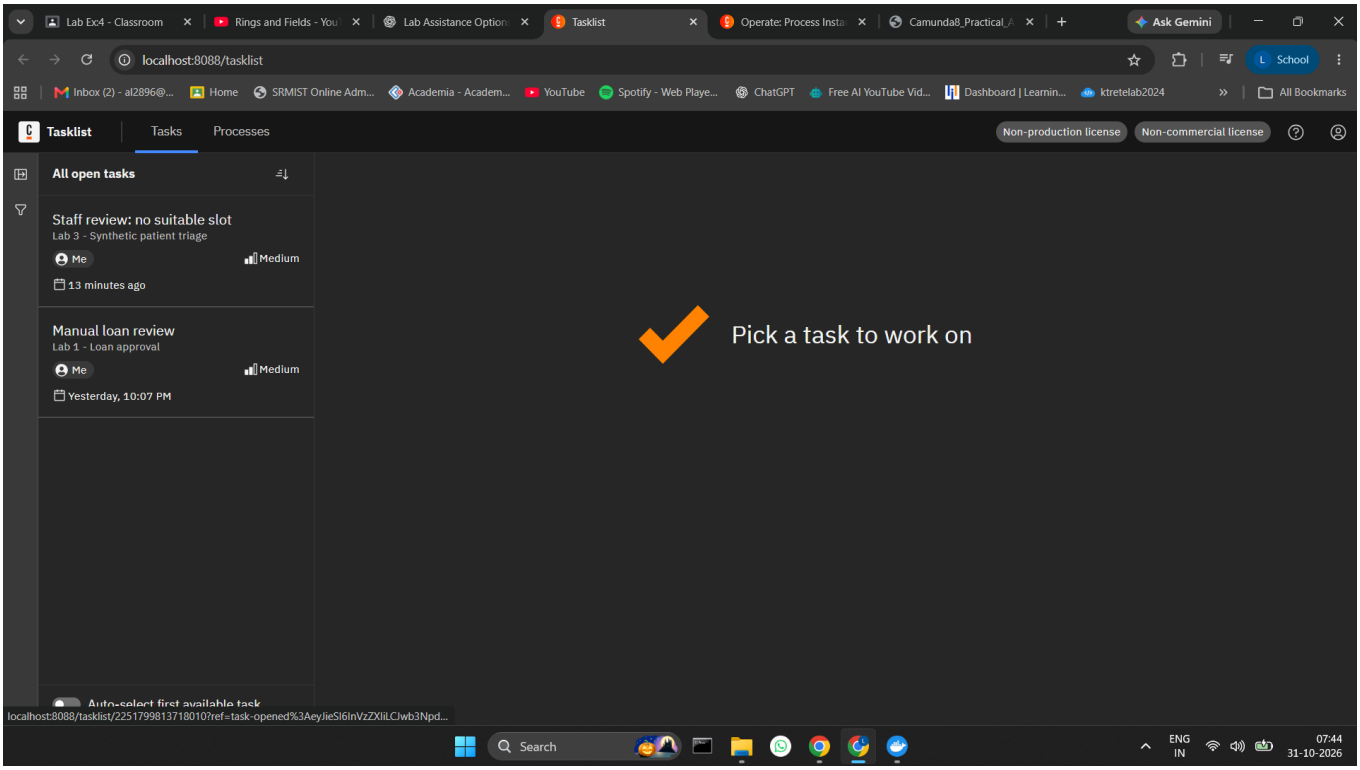
Completed urgent route; staffActionRecorded true and review notes.



Full-resolution evidence: evidence/runtime/H3-completed.png

H4-task-list

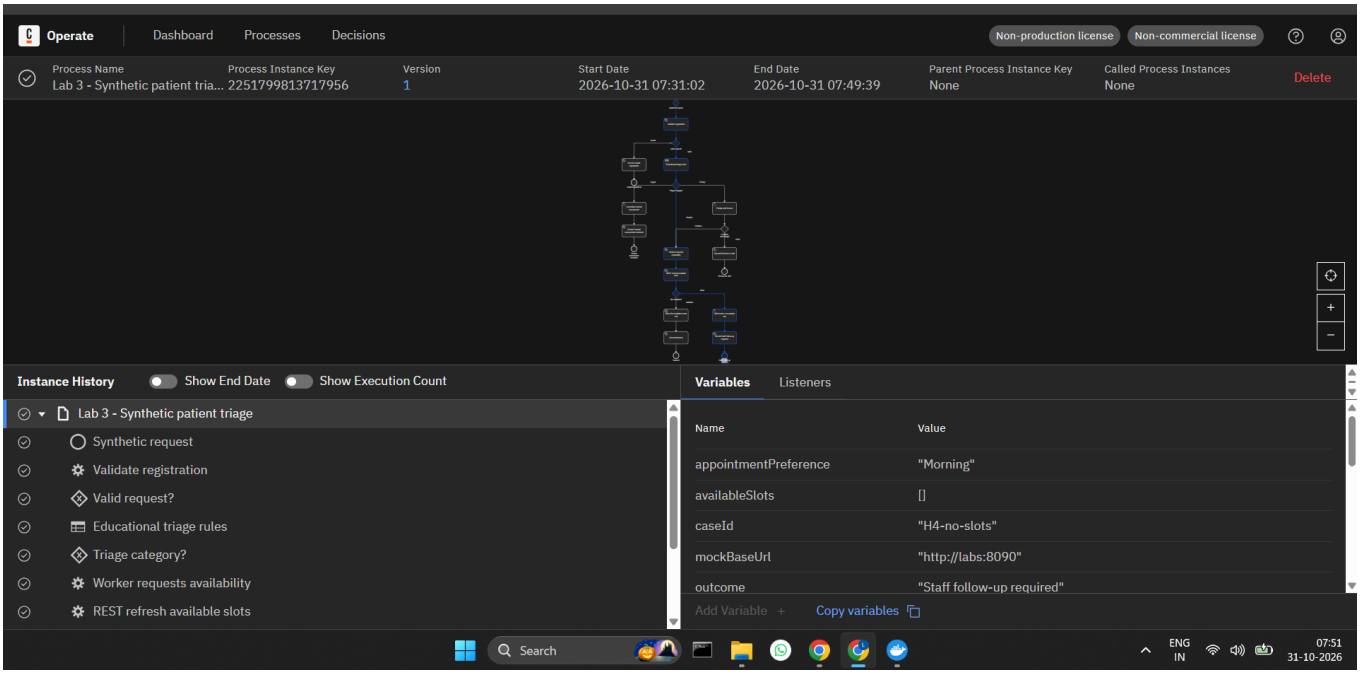
No-slot staff review task listed in Tasklist.



Full-resolution evidence: evidence/runtime/H4-task-list.png

H4-completed

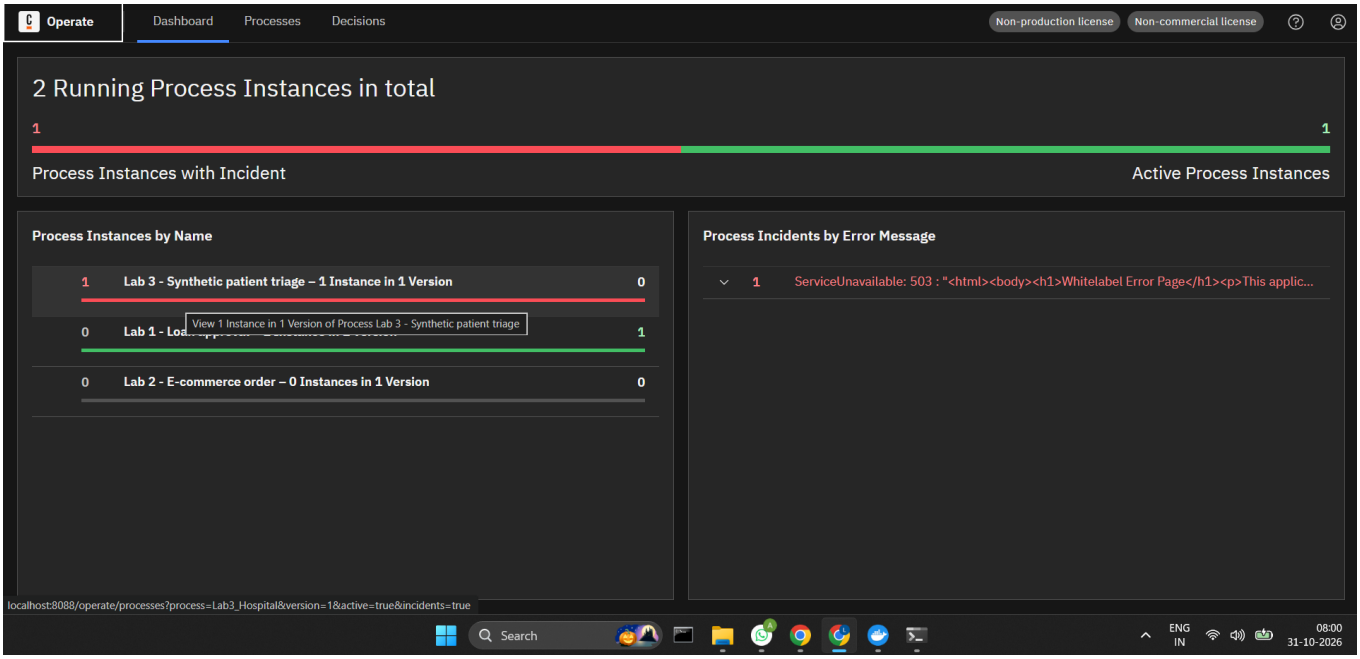
Completed no-slot case; Staff follow-up required outcome.



Full-resolution evidence: evidence/runtime/H4-completed.png

H5-incident-dashboard

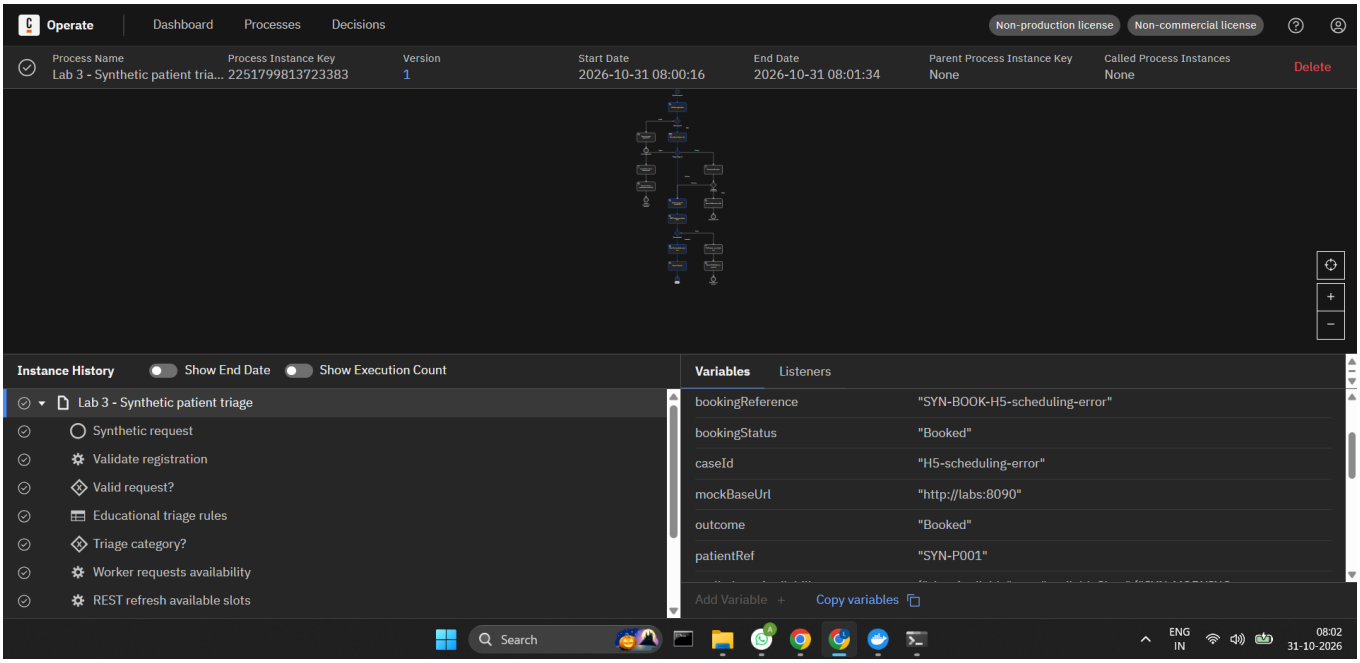
Lab 3 dashboard shows one HTTP 503 incident. No process instance key is visible, so this image alone cannot establish same-instance linkage.



Full-resolution evidence: evidence/runtime/H5-incident-dashboard.png

H5-recovered

H5 instance 2251799813723383 completed at 08:01:34; bookingStatus and outcome Booked.



Full-resolution evidence: evidence/runtime/H5-recovered.png